

Topic 6

Microscopic World II

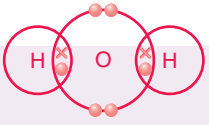
Chemistry in Context

Ice fishing Ice fishing is the practice of catching fish with lines and fish hooks through an opening in the ice on a frozen lake. The fact that we are able to practise ice fishing has something to do with the structure of the water molecules.



When liquid water is cooled, it contracts like we would expect until a temperature of approximately 4 °C is reached. After that, it expands until it reaches 0 °C (freezing point), and then when it freezes and becomes ice, it expands by approximately 9%. Ice has an open structure with large spaces between its molecules. It is this open structure that causes ice to be less dense than liquid water. That is why ice floats on water.

One consequence of this is that lakes and rivers freeze from top to bottom, allowing aquatic lives to survive even when the surface of a lake or river has frozen. Hence we are able to practise ice fishing.



Do you recall?

1 Draw an electron diagram of water, showing electrons in the outermost shells only.

2 What is the type of bonding present in a water molecule?
Covalent bond

3 According to the bonding suggested in Q2, what type of structure does water have?
Simple molecular structure

You are going to find out the type of intermolecular forces that lead to the open structure of ice.

Unit

23 Shapes of molecules

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Covalent molecules with non-octet structures
- Shapes of molecules
- Structure of buckminsterfullerene (C_{60})

23.1 Covalent molecules with non-octet structures

Fig. 23.1 shows the electron diagrams of molecules of methane and ammonia. In Topic 2 Microscopic World I, we notice that non-metals in the second period of the periodic table (such as carbon and nitrogen) form stable molecules when they have eight electrons (i.e. an octet structure) in their outermost shells.



methane ammonia

Fig. 23.1 Electron diagrams of molecules of methane and ammonia

However, some elements do not follow this rule. We are going to look at some examples. N1

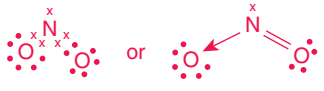
Boron trifluoride molecule (BF_3)

Boron is in Group III of the periodic table. It can form compounds having less than eight electrons in the outermost shell of the boron atom.

N1 Molecules to which the 'octet rule' is NOT applicable

Questions often ask students to identify molecules to which the 'octet rule' is NOT applicable. The octet rule suggests that atoms become stable by having eight electrons in their outermost shells (or two electrons in the case of some smaller atoms). An atom attains a stable electronic arrangement by sharing or transfer of electrons.

- Some compounds exist as radicals (a radical is a species that contains at least one unpaired electron), e.g. NO_2.



- Some molecules contain atoms with less than 8 outermost shell electrons, e.g. BeH_2, BF_3.
- Some molecules contain atoms with more than 8 outermost shell electrons, e.g. PCl_5, SF_6.

A boron atom has three electrons in the outermost shell. Boron reacts with halogens to form a class of compounds having the general formula BX_3, where X is a halogen atom. In a **boron trifluoride** molecule, there are only 6 electrons in the outermost shell of the boron atom (Fig. 23.2). Boron trifluoride is a molecule that does not have an octet structure.

DSE 2014 Paper 1A Q22

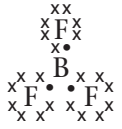


Fig. 23.2 In a BF_3 molecule, there are only 6 electrons in the outermost shell of the boron atom

Phosphorus pentachloride molecule (PCl_5)

N2 Nitrogen forms only one chloride, NCl_3, while phosphorus forms two chlorides, PCl_3 and PCl_5.

Elements in the second period of the periodic table cannot form N2 compounds with more than eight electrons in the outermost shells of their atoms. However, elements in and beyond the third period form some compounds with more than eight electrons in the outermost shells of their atoms. Phosphorus is an example.

A phosphorus atom has an electronic arrangement of 2,8,5. In a **phosphorus pentachloride** molecule, each of the five outermost shell electrons of the phosphorus atom forms a covalent bond with an electron of a chlorine atom. Thus, there are 10 electrons in the outermost shell of the phosphorus atom (Fig. 23.3). Phosphorus pentachloride is a molecule with an extended octet structure.

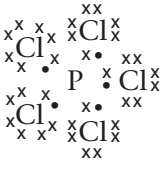


Fig. 23.3 In a PCl_5 molecule, there are 10 electrons in the outermost shell of the phosphorus atom

boron trifluoride 三氟化硼　　phosphorus pentachloride 五氯化磷

Sulphur hexafluoride molecule (SF_6)

N3 Electron diagram of a SF_4 molecule:



Sulphur hexafluoride (SF_6) is another molecule with an extended N3 octet structure.

A sulphur atom has an electronic arrangement of 2,8,6. In a sulphur hexafluoride molecule, each of the six outermost shell electrons of the sulphur atom forms a covalent bond with an electron of a fluorine atom. Thus, there are 12 electrons in the outermost shell of the sulphur atom (Fig. 23.4).

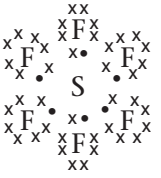


Fig. 23.4 In a SF_6 molecule, there are 12 electrons in the outermost shell of the sulphur atom

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discuss whether you agree with the following statement.

'Both oxygen and sulphur can form hexafluoride, i.e. OF_6 and SF_6.'

ASL 2011 Paper 1 Q2(b)
AL 2011 Paper 1 Q1(b)(ii)

Not agree.

Oxygen is in the second period of the periodic table. It cannot form compounds with more than 8 electrons in the outermost shell of its atom.

Practice 23.1

1 Consider the following species:

CS_2, SCl_4

For each species,

a) draw an electron diagram of the molecule;  

b) decide whether the species follows the 'octet rule'.

CS_2 follows the 'octet rule'.
SF_4 does not follow the 'octet rule'.

Continued on next page

sulphur hexafluoride 六氟化硫

2 Part of the electron diagram of nitrogen dioxide, showing the electrons around the nitrogen atom, is shown below:



a) Complete the electron diagram above by adding electrons around the oxygen atoms.

b) Describe the type of bond shown between the nitrogen and the right-hand oxygen atom.
Dative covalent bond

c) Decide whether nitrogen dioxide follows the 'octet rule'.
No.

Reference Website
Refer to the following website for animated spinning models of methane, ammonia and water molecules:
http://winter.group.shef.ac.uk/vsepr/examples.html
(accessed July 2014)

23.2 Shapes of molecules of methane, ammonia and water

Electron pair repulsion theory

ASL 2006 Q14

Because all electrons have the same (negative) charge, they repel each other when they are close together. So, a pair of electrons in any of the bonds surrounding the central atom in a molecule will repel other electron pairs. Thus, the pairs of electrons will stay as far apart as possible to minimize repulsion.

Shape of a methane molecule

Fig. 23.5a shows an electron diagram of a methane molecule. There are four pairs of electrons in the outermost shell of the central carbon atom — four bond pairs of electrons. The repulsive forces of each bond pair of electrons result in a **tetrahedral** shape. Thus, the methane molecule has a tetrahedral shape with H atoms at the corners and the C atom in the centre (Fig. 23.5b–c).

DSE 2012 Paper 1A Q12

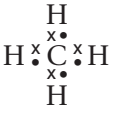


Fig. 23.5a An electron diagram of a methane molecule

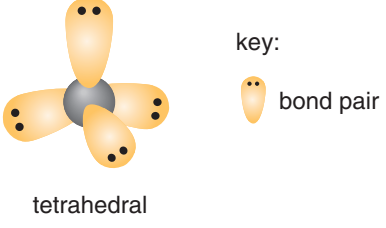


Fig. 23.5b The arrangement of electron pairs around the central carbon atom in a methane molecule

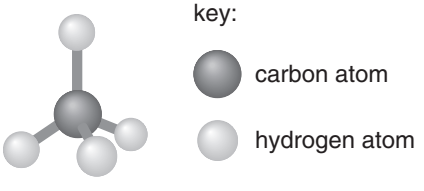


Fig. 23.5c A 'ball-and-stick' model of a methane molecule

electron pair repulsion theory 電子對相斥學說　　tetrahedral 四面體的

Shape of an ammonia molecule

Fig. 23.6a shows an electron diagram of an ammonia molecule. An ammonia molecule has one lone pair and three bond pairs of electrons in the outermost shell of the nitrogen atom. These electron pairs repel to get as far apart as possible. The four electron pairs will adopt a tetrahedral arrangement (Fig. 23.6b).

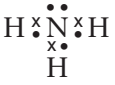


Fig. 23.6a An electron diagram of an ammonia molecule

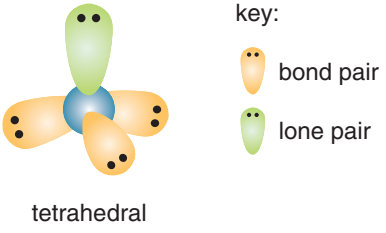


Fig. 23.6b The arrangement of electron pairs around the central nitrogen atom in an ammonia molecule

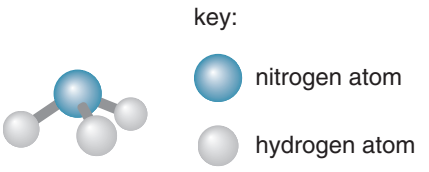


Fig. 23.6c A 'ball-and-stick' model of an ammonia molecule

DSE 2012 Paper 1A Q12
ASL 2012 Paper 1 Q3(d)
AL 2012 Paper 1 Q3(b)(iv)

Notice that the shape of a molecule is determined only by the arrangement of atoms. Thus, the ammonia molecule has a **trigonal pyramidal** shape (Fig. 23.6c).

ASL 2009 Paper 1 Q1(a)
AL 2009 Paper 1 Q1(a)(i)

Shape of a water molecule

Fig. 23.7a shows an electron diagram of a water molecule. A water molecule has two lone pairs and two bond pairs of electrons in the outermost shell of the oxygen atom. These electron pairs repel to get as far apart as possible. The four electron pairs will adopt a tetrahedral arrangement (Fig. 23.7b).



Fig. 23.7a An electron diagram of a water molecule

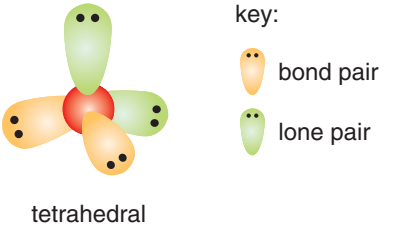


Fig. 23.7b The arrangement of electron pairs around the central oxygen atom in a water molecule

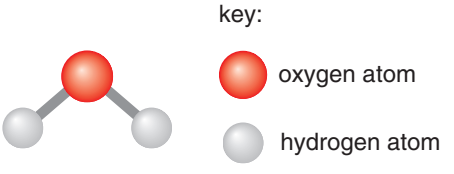


Fig. 23.7c A 'ball-and-stick' model of a water molecule

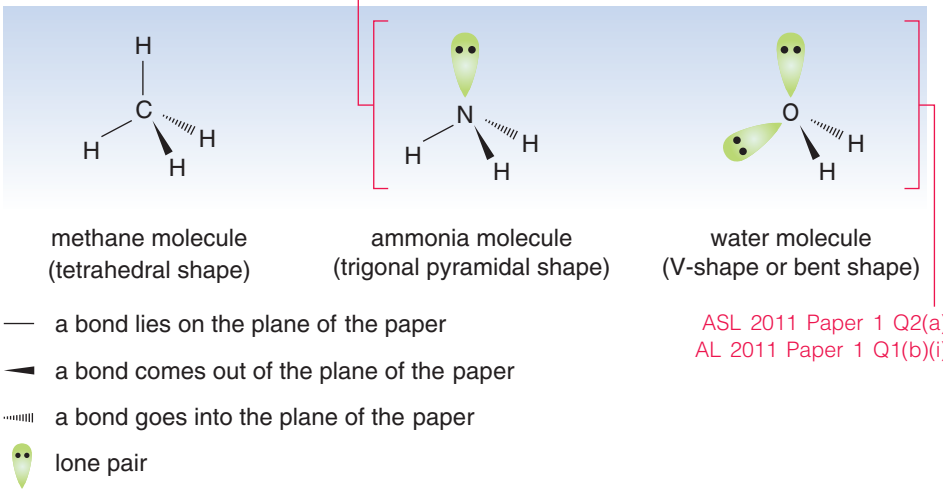
The shape of a molecule is determined only by the arrangement of atoms. The water molecule has a **V-shape** or **bent** shape (Fig. 23.7c).

trigonal pyramidal 三角錐體的　　V-shape V形的　　bent 角形的

Drawing 3-D structures of molecules

For the convenience of communication, chemists have developed a system for drawing the 3-D structures of molecules on paper. Fig. 23.8 shows drawings of the 3-D structures of molecules of methane, ammonia and water.

DSE 2013 Paper 1B Q2(a)
ASL 2010 Paper 1 Q4(b)



ASL 2011 Paper 1 Q2(a)
AL 2011 Paper 1 Q1(b)(i)

Fig. 23.8 The 3-D representations of methane, ammonia and water molecules

Practice 23.2

Refer to page T9.

Consider the following molecules:

1 SiF_4;

2 SCl_2; and

3 PH_3.

For each molecule,

a) draw an electron diagram;

b) draw a 3-D structure, showing all electron pairs in the outermost shell of the central atom;

c) state the shape of the molecule.

23.3 Shapes of some other molecules

Shape of a beryllium chloride molecule ($BeCl_2$)

In a beryllium chloride molecule, there are two bond pairs of electrons in the outermost shell of the central beryllium atom (Fig. 23.9a). The two electron pairs repel each other. They must be at opposite ends of a straight line in order to be as far apart as possible. Thus, the molecule is **linear** in shape (Fig. 23.9b). Fig. 23.9c shows a 'ball-and-stick' model of a beryllium chloride molecule.

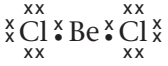


Fig. 23.9a An electron diagram of a beryllium chloride molecule

Fig. 23.9b A beryllium chloride molecule is linear in shape



Fig. 23.9c A 'ball-and-stick' model of a beryllium chloride molecule

DSE 2012 Paper 1A Q12
ASL 2012 Paper 1 Q3(d)
AL 2012 Paper 1 Q3(b)(iv)

Shape of a boron trifluoride molecule (BF_3) N4

In a boron trifluoride molecule, there are three bond pairs of electrons in the outermost shell of the central boron atom (Fig. 23.10a). These electron pairs repel to get as far apart as possible. The shape that puts the three electron pairs furthest apart is **trigonal planar**.

The boron trifluoride molecule is flat and it is trigonal planar in shape (Fig. 23.10b). Fig. 23.10c shows a 'ball-and-stick' model of a boron trifluoride molecule.

N4 Examination questions often ask about the shape of a BF_3 molecule.

N5 The 3-D representation of a BF_3 molecule:



N6 Refer to page T9.

ASL 2009 Paper 1 Q1(a)
AL 2009 Paper 1 Q1(a)(i)

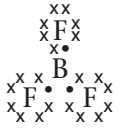


Fig. 23.10a An electron diagram of a boron trifluoride molecule

DSE 2013 Paper 1B Q2(a)
AL 2007 Paper 1 Q3(c)(iii)

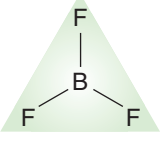


Fig. 23.10b A boron trifluoride molecule is trigonal planar in shape N5

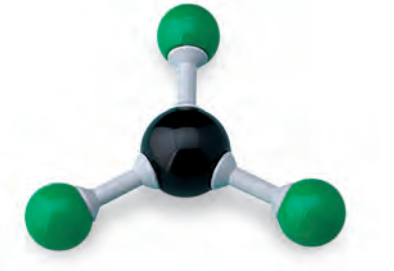


Fig. 23.10c A 'ball-and-stick' model of a boron trifluoride molecule N6

ASL 2009 Paper 1 Q1(a)
AL 2009 Paper 1 Q1(a)(i)

linear 線形的　　trigonal planar 平面三角的

Shape of a phosphorus pentachloride molecule (PCl_5)

DSE 2012 Paper 1A Q12

In a phosphorus pentachloride molecule, there are five bond pairs of electrons in the outermost shell of the central phosphorus atom (Fig. 23.11a). These electron pairs repel to get as far apart as possible. The shape that puts the five electron pairs furthest apart is **trigonal bipyramidal**. Thus, the molecule is trigonal bipyramidal in shape (Fig. 23.11b). Fig. 23.11c shows a 'ball-and-stick' model of a phosphorus pentachloride molecule.

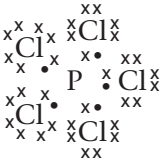
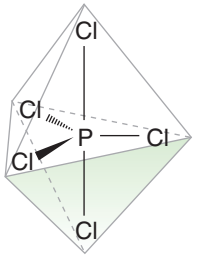


Fig. 23.11a An electron diagram of a phosphorus pentachloride molecule



ASL 2010 Paper 1 Q4(b)

Fig. 23.11b A phosphorus pentachloride molecule is trigonal bipyramidal in shape



Fig. 23.11c A 'ball-and-stick' model of a phosphorus pentachloride molecule

23.1
Building 'balloon' molecules.

Shape of a sulphur hexafluoride molecule (SF_6)

In a sulphur hexafluoride molecule, there are six bond pairs of electrons in the outermost shell of the central sulphur atom (Fig. 23.12a). These electron pairs repel to get as far apart as possible. The shape that puts the six electron pairs furthest apart is **octahedral**. Thus, the molecule is octahedral in shape (Fig. 23.12b). Fig. 23.12c shows a 'ball-and-stick' model of a sulphur hexafluoride molecule.

23.2
Building ball-and-stick models of molecules.

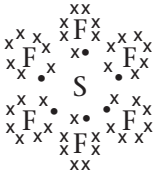
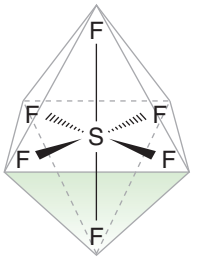


Fig. 23.12a An electron diagram of a sulphur hexafluoride molecule



ASL 2011 Paper 1 Q2(a)
AL 2011 Paper 1 Q1(b)(i)

Fig. 23.12b A sulphur hexafluoride molecule is octahedral in shape



Fig. 23.12c A 'ball-and-stick' model of a sulphur hexafluoride molecule

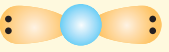
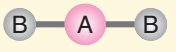
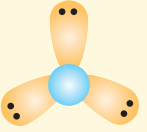
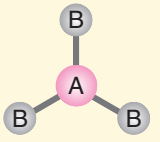
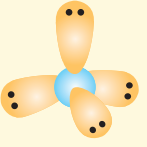
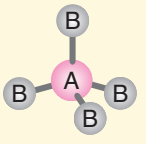
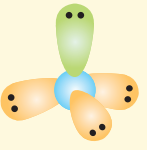
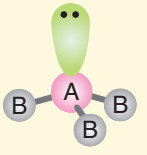
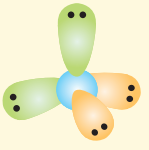
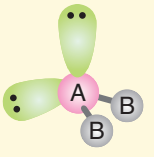
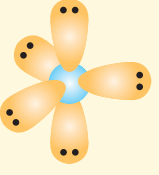
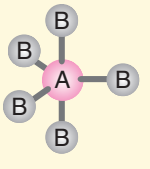
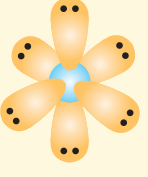
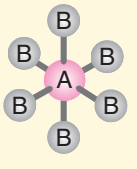
trigonal bipyramidal 三角雙錐體的　　octahedral 八面體的

N7 Refer to page T10.

Our discussion concerning shapes of molecules is summarized in Table 23.1.

Table 23.1

A summary of shapes of molecules N7

In the outermost shell of the central atom,			Arrangement of electron pairs	Molecular shape (determined by arrangement of atoms)	Example(s)
number of bond pairs	number of lone pairs	total number of electron pairs	key: bond pair, lone pair		
2	0	2	 linear	 linear	$BeCl_2$
3	0	3	 trigonal planar	 trigonal planar	BF_3
4	0	4	 tetrahedral	 tetrahedral	CH_4 $CHCl_3$ $SiCl_4$
3	1		 tetrahedral	 trigonal pyramidal	NH_3 PH_3
2	2		 tetrahedral	 V-shape or bent shape	H_2O H_2S
5	0	5	 trigonal bipyramidal	 trigonal bipyramidal	PCl_5 PBr_5
6	0	6	 octahedral	 octahedral	SF_6 SCl_6

23.4 Shapes of molecules with multiple bonds

Predictions about shapes of molecules can be extended to cover molecules with multiple bonds. One general rule governing the use of the electron pair repulsion theory is that multiple bonds can be counted as single bonds.

Shape of a carbon dioxide molecule (CO_2)

Fig. 23.13a shows an electron diagram of a carbon dioxide molecule. When using the electron pair repulsion theory, we can count double bonds as single bonds. Thus, we can view the carbon atom as having two pairs of electrons in its outermost shell. The two electron pairs repel each other. They must be at opposite ends of a straight line in order to be as far apart as possible. Thus, a carbon dioxide molecule is linear in shape (Fig. 23.13b). Fig. 23.13c shows a 'ball-and-stick' model of a carbon dioxide molecule.

DSE 2014 Paper 1B Q1(a)(iii)

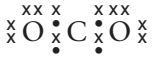


Fig. 23.13a An electron diagram of a carbon dioxide molecule

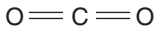


Fig. 23.13b A carbon dioxide molecule is linear in shape



Fig. 23.13c A 'ball-and-stick' model of a carbon dioxide molecule

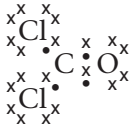
Example 23.1

Q Consider the molecule $COCl_2$.

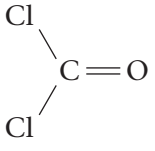
a) Draw an electron diagram for the molecule.

b) Draw a 3-D structure and state the shape of the molecule.

A a)



b)



When using the electron pair repulsion theory, double bonds can be counted as single bonds. Thus, we can view the carbon atom as having three pairs of electrons in its outermost shell.

These electron pairs repel to get as far apart as possible. The shape that puts the three electron pairs furthest apart is trigonal planar. Thus, the $COCl_2$ molecule is trigonal planar in shape.

Practice 23.3 Refer to page T11.

Below are the electron diagrams for two molecules (HCN and HCHO):



Describe and explain the shape of each molecule.

Do you know

The structure of C_{60}

Radioastronomy had revealed that long chains of carbon atoms existed in outer space. When scientists tried to re-create the conditions that might account for the carbon chains, they found a new molecule formed by chance. The molecule was made of 60 carbon atoms, but it was unclear how the atoms were arranged within each molecule.

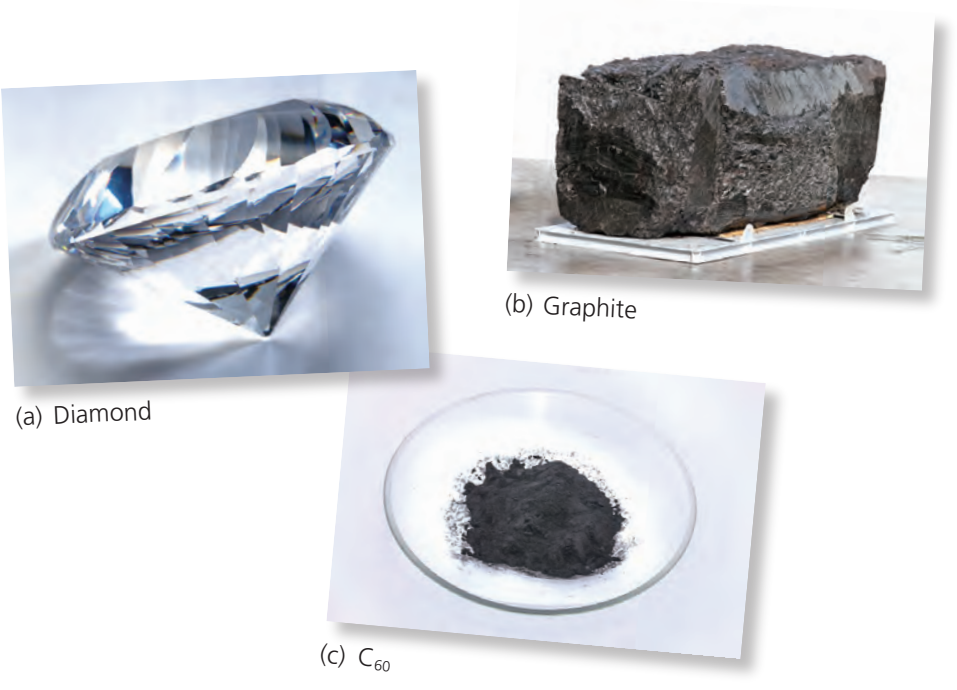
Professor Harry Kroto◂ solved the problem by suggesting a structure of hexagons and pentagons arranged in a sphere. The structure of C_{60} was thus determined.

◂ Professor Harry Kroto shared the 1996 Nobel Prize in Chemistry with two other scientists.

23.5 Molecular crystals of buckminsterfullerene (C_{60})

Structure of buckminsterfullerene (C_{60})

The two carbon allotropes, diamond and graphite (Figs. 23.14a–b), have been known for centuries. In 1985, a new allotrope of carbon (C_{60}) (Fig. 23.14c) was discovered.



(a) Diamond

(b) Graphite

(c) C_{60}

Fig. 23.14 Allotropic forms of carbon and their appearances

23.3

Building a paper model of a C_{60} molecule.

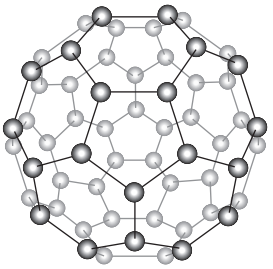


Fig. 23.15 Structure of a C_{60} molecule



Fig. 23.16 A geodesic dome with a framework similar to the arrangement of carbon atoms in a C_{60} molecule

N8 C_{60} molecule is non-polar.

N9 Fullerenes are soluble in some organic solvents because the van der Waals' forces between molecules of fullerenes are of similar strength as those in organic solvents.

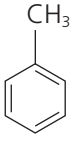
A C_{60} molecule consists of 60 carbon atoms which form the shape N8 of a ball like a soccer ball with a carbon atom at each corner of the 20 hexagons and 12 pentagons (Fig. 23.15). This form of carbon was formally named **buckminsterfullerene** after the American architect Buckminster Fuller, who was famous for designing a large geodesic dome which looked similar to the molecular structure of C_{60} (Fig. 23.16). Many other molecules of carbon with ball shapes called **fullerenes** have since been isolated, including C_{70}, C_{76} and C_{84}. Spherical fullerenes have become known as '**buckyballs**'.

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Properties of buckminsterfullerene

Good solvents for fullerenes include toluene◂ and carbon disulphide. N9 Buckminsterfullerene dissolves in toluene to give a pink solution. It is as soft as graphite and is a semiconductor. Its natural form is a very fine, black crystalline powder.

Toluene is a non-aqueous solvent with the following structure:



At room temperature, molecules in solid C_{60} are closely packed and bound by weak van der Waals' forces◂. In a molecule of C_{60}, each carbon atom forms three single covalent bonds with other carbon atoms.

We will further discuss van der Waals' forces in Unit 24.

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Buckminsterfullerene was discovered so recently that its properties have not been fully established. Most of the reactions thus far reported for C_{60} involve addition◂ to the core structure. These reactions include hydrogenation and bromination.

We will further discuss addition reactions in Topic 7 Fossil Fuels and Carbon Compounds.

Buckminsterfullerene also forms compounds with electron donating atoms readily, the most common examples being alkali metals. For example, it reacts with potassium under certain conditions to give an ionic compound with the formula K_3C_{60}, which is a *superconductor* when cooled to –255 °C.

buckminsterfullerene 布克碳　fullerene 富勒烯　buckyball 布克球　superconductor 超導體

Applications of fullerenes

There has been much research on practical applications of fullerenes. With its special properties, a fullerene can be manipulated to make materials never known before.

- Its stability and bonding properties make it a suitable lubricant, which can be bettered by adding fluorine atoms (as is done in Teflon) to the outside of the molecule.

- Its molecule has enough space inside to hold any known element (Fig. 23.17). This fits for medicinal purposes such as drug delivery systems, targeted cancer therapies and chemical *tracers*.

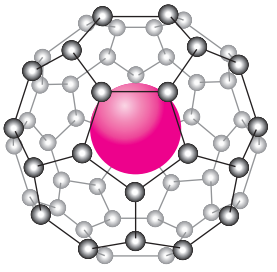


Fig. 23.17 Buckminsterfullerene with a trapped element

- Buckminsterfullerene becomes harder than diamond when compressed to less than 70% of its volume. Hence it can be used as a cutting tool or hardening agent.

- Buckminsterfullerene stops light effectively. Very small quantities of it can stop several laser pulses completely. A fullerene can also be made as a safe storage molecule for radioactive waste.

Interest in the fullerenes has led to the discovery of a related group of carbon structures referred to as **nanotubes**. Nanotubes may be viewed as rolled up segments of graphite (Fig. 23.18).

We will further discuss nanomaterials in Topic 14 Materials Chemistry.

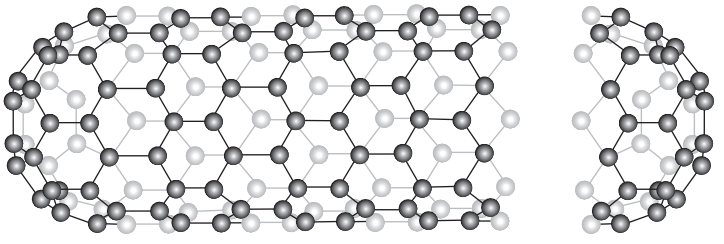


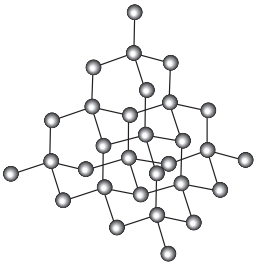
Fig. 23.18 Structure of a carbon nanotube (rolled up segment of graphite in the form of a cylinder with both ends often closed)

tracer 示蹤物　　nanotube 納米管

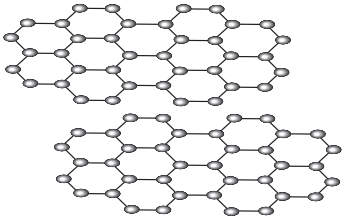
Practice 23.4

Refer to page T12.

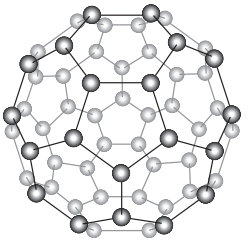
The figures below show the structural models of three different allotropic forms of carbon — diamond, graphite and buckminsterfullerene.



diamond



graphite



buckminsterfullerene

1 Why is C_{60} soft and slippery?

2 Why is buckminsterfullerene a 'powdery' solid rather than being composed of large-scale 'chunks'?

3 Suggest why buckminsterfullerene dissolves in petrol but diamond and graphite do not.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Carbon nanotubes — new material with superior properties

Carbon nanotubes (CNTs) are a new form of carbon, first identified in 1991 by Sumio Iijima, a Japanese scientist.

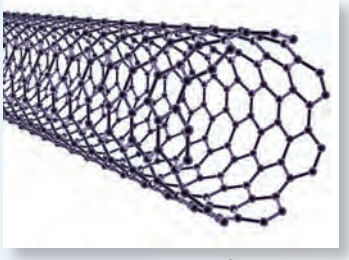


Fig. 23.19 Model of a carbon nanotube

Carbon nanotubes (Fig. 23.19) possess many remarkable properties. For example, it is estimated that they are 100 times stronger than steel at only one-sixth of the mass, they conduct electricity better than copper and transmit heat better than diamond.

Possible applications for carbon nanotubes include flat-screen LCD screens, hydrogen storage, drug delivery, catalysts and memory-storage devices.

Carbon nanotubes for cleaning polluted water

Scientists found that by using filters made of carbon nanotubes, pollutants could be removed more effectively from contaminated water as compared to common charcoal filters. Carbon nanotubes have a very large surface area (e.g. 500 m^2 per gram of nanotube) that gives them a high capacity to retain pollutants. However, there are still many health and environmental questions to answer before such filters find their way into municipal water treatment plants.

Questions

1 Suggest why some experts and consumers may worry about the effects of using filters made of carbon nanotubes in municipal water treatment plants.

2 What actions may be taken to reduce the concerns of experts and consumers?

Refer to page T12.

Have you mastered?

Key terms

boron trifluoride 三氟化硼 3 phosphorus pentachloride 五氯化磷 3

sulphur hexafluoride 六氟化硫 4 electron pair repulsion theory 電子對相斥學說 5

tetrahedral 四面體的 5 trigonal pyramidal 三角錐體的 6 V-shape V形的 6 bent 角形的 6

linear 線形的 8 trigonal planar 平面三角的 8 trigonal bipyramidal 三角雙錐體的 9

octahedral 八面體的 9 buckminsterfullerene 布克碳 13 fullerene 富勒烯 13

buckyball 布克球 13 nanotube 納米管 14

Checklist

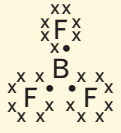
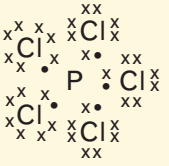
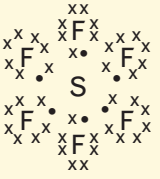
After studying this unit, you should be able to

- ☐ draw the electron diagrams of some non-octet molecules;
- ☐ describe and draw 3-D diagrams to represent shapes of molcules with central atoms obeying the octet rule;
- ☐ describe and draw 3-D diagrams to represent shapes of molecules with central atoms not obeying the octet rule and with no lone pair of electrons;
- ☐ describe the structure of buckminsterfullerene;
- ☐ state and explain the properties of buckminsterfullerene in terms of its structure and bonding.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

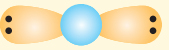
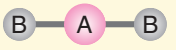
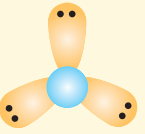
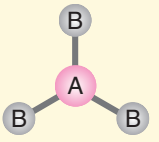
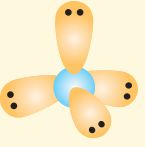
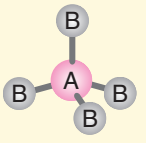
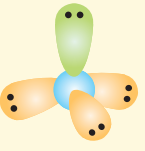
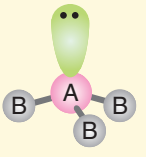
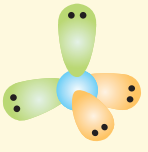
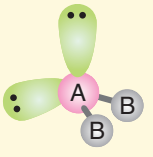
Summary

1 Examples of covalent molecules with non-octet structures are as follows:

Molecule	Electron diagram
BF_3	
PCl_5	
SF_6	

Have you mastered?

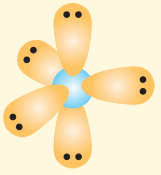
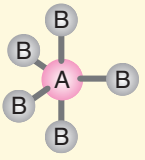
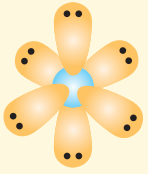
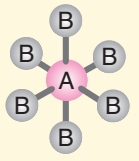
2 The following table summarizes the relationship between the arrangement of electron pairs around the central atom of a molecule and the shape of the molecule.

Total number of electron pairs in the outermost shell of the central atom of a molecule	Arrangement of electron pairs	Shape of the molecule	Examples
2	 linear	 linear	$BeCl_2$ CO_2
3	 trigonal planar	 trigonal planar	BF_3 $COCl_2$
4	 tetrahedral	 tetrahedral	CH_4 $CHCl_3$ $SiCl_4$
	 tetrahedral	 trigonal pyramidal	NH_3 PH_3
	 tetrahedral	 V-shape or bent shape	H_2O H_2S

key: bond pair lone pair

Continued on next page

When using the electron pair repulsion theory, count multiple bonds as single bonds.

Total number of electron pairs in the outermost shell of the central atom of a molecule	Arrangement of electron pairs	Shape of the molecule	Examples
5	 trigonal bipyramidal	 trigonal bipyramidal	PCl_5 PBr_5
6	 octahedral	 octahedral	SF_6 SCl_6

3 a) A C_{60} molecule consists of 60 carbon atoms which form the shape of a ball like a soccer ball with a carbon atom at each corner of the 20 hexagons and 12 pentagons.

b) C_{60} is as soft as graphite and is a semiconductor.

c) At room temperature, molecules in solid C_{60} are closely packed and bound by weak van der Waals' forces.

d) Interest in the fullerenes has led to the discovery of a related group of carbon structures referred to as nanotubes.

Have you mastered?

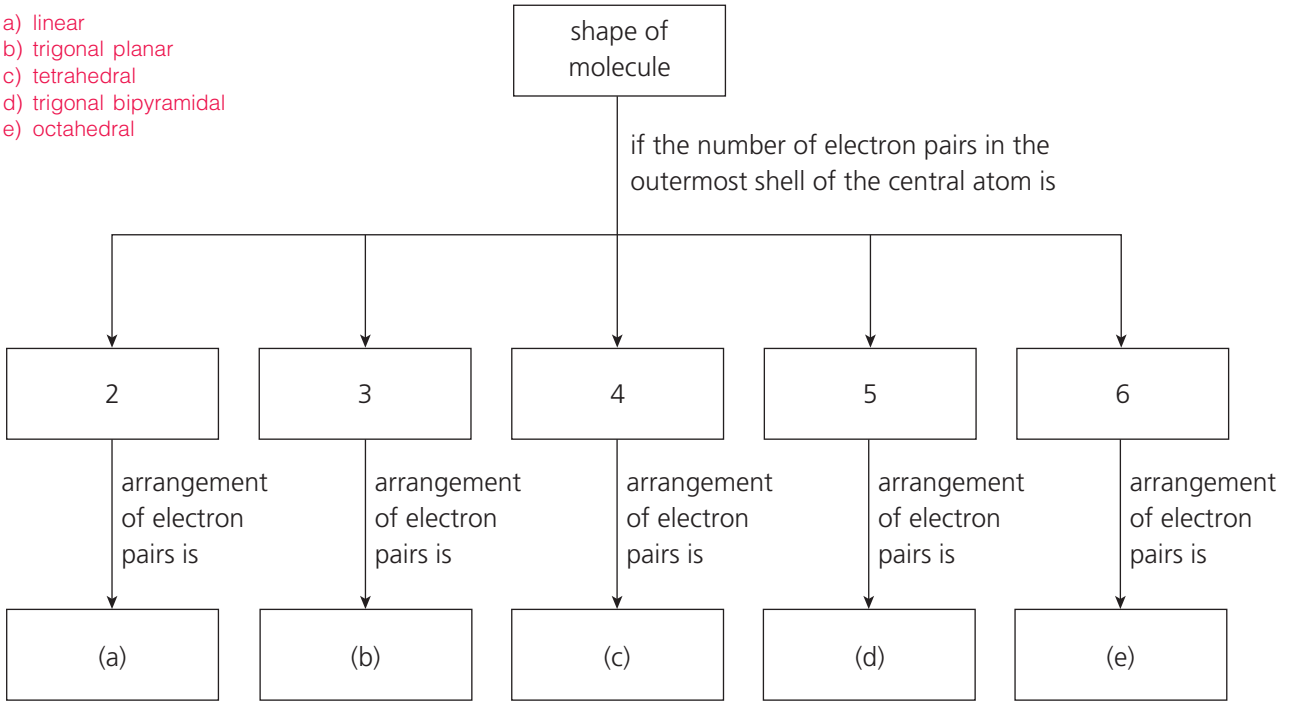
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

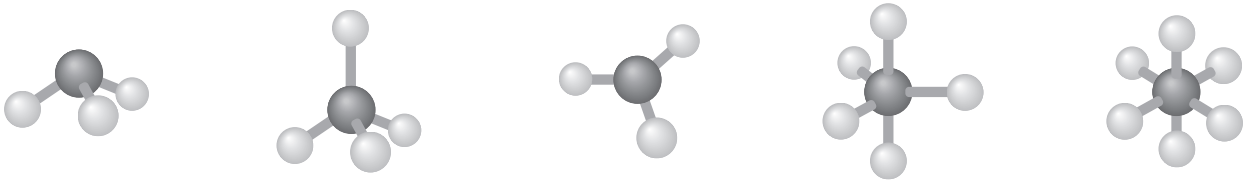
Part I Knowledge and understanding

1 Complete the following concept map.

a) linear
b) trigonal planar
c) tetrahedral
d) trigonal bipyramidal
e) octahedral



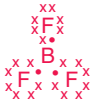
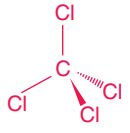
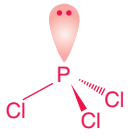
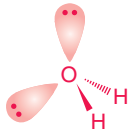
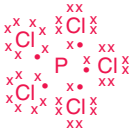
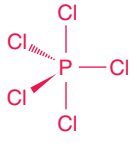
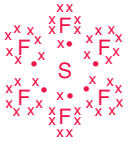
2 Refer to the following molecular models.



(a) Trigonal pyramidal
(b) Tetrahedral
(c) Trigonal planar
(d) Trigonal bipyramidal
(e) Octahedral

What is the shape of each molecule?

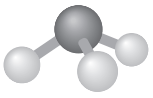
3 Complete the following table.

Molecule	Electron diagram (showing electrons in the *outermost shells* only)	Number of bond pairs in the outermost shell of the central atom	Number of lone pairs in the outermost shell of the central atom	Three-dimensional structure (showing all electron pairs in the outermost shell of the central atom)	Shape of molecule
$BeCl_2$		2	0		linear
BF_3		3	0		trigonal planar
CCl_4		4	0		tetrahedral
PCl_3		3	1		trigonal pyramidal
H_2O		2	2		V-shape or bent shape
PCl_5		5	0		trigonal bipyramidal
SF_6		6	0		octahedral

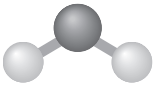
4 Atoms react together to form molecules in the dense gas clouds in space. Molecules of H_2S, NH_3 and OCS (similar in structure to CO_2) have been detected.

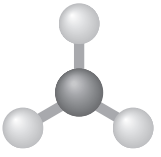
a) Draw an electron diagram of each molecule.

b) Decide which of the following models represents the shape of each molecule.

(i) 

(ii) 

(iii) 

(iv) 

Molecule	Electron diagram	Model
H_2S		(iii)
NH_3		(i)
OCS		(ii)

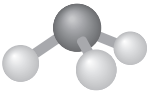
Part II Multiple choice questions

5 Which of the following molecules has an electron diagram with exactly one lone pair of electrons in the outermost shell of the central atom?
Refer to page T13 for explanation.
A CCl_4
B H_2O
C PH_3
D SF_6

6 Which of the species shown below does NOT follow the 'octet rule'?

A Na_2O
B MgO
C PCl_3
D SCl_4

(HKDSE, Practice paper 1A, 2012, 2)

7 

Which of the following molecules could this model represent?
Refer to page T13 for explanation.
A Boron trifluoride
B Hydrogen sulphide
C Methane
D Nitrogen trichloride

Directions: Questions 8 and 9 concern the shapes of molecules.

A Linear
B Trigonal planar
C Pyramidal
D Tetrahedral

8 Select from A to D the shape of boron trichloride, BCl_3. B
Refer to page T13 for explanation.

9 Select from A to D the shape of carbon dioxide, CO_2. A
Refer to page T14 for explanation.
(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 1(a), (c))

10 Which of the following species does NOT exist?

A NCl_3
B NCl_5
C SF_2
D SF_6

NCl_5 does NOT exist. Nitrogen cannot form compounds with more than 8 electrons in the outermost shell of its atom.

11 What is the shape of a phosphorus pentabromide molecule?
Refer to page T14 for explanation.
A Octahedral
B Planar
C Trigonal bipyramidal
D V-shape

12 Which of the following molecules has a trigonal planar shape?

Refer to page T14 for explanation.

(A) $COCl_2$
B H_2S
C PH_3
D SCl_2

13 Which of the following molecules have a linear shape?

Refer to page T14 for explanation.

(1) $BeCl_2$
(2) CS_2
(3) OF_2

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

14 Which of the following molecules have a trigonal pyramidal shape?

Refer to page T14 for explanation.

(1) BF_3
(2) NH_3
(3) PCl_3

A (1) and (2) only
B (1) and (3) only
(C) (2) and (3) only
D (1), (2) and (3)

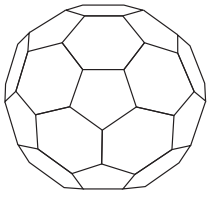
15 In which of the following pairs do the molecules have a similar shape?

(1) $BeCl_2$ and HCN
(2) BCl_3 and NF_3
(3) CH_4 and $SiBr_4$

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(1) $BeCl_2$ and HCN molecules have a linear shape.
(2) A BCl_3 molecule has a trigonal planar shape while a NF_3 molecule has a trigonal pyramidal shape.
(3) CH_4 and $SiBr_4$ molecules have a tetrahedral shape.

16 Buckminsterfullerene has the formula C_{60}. Its structure is shown below.



Which of the following statements are correct?

(1) The melting point of buckminsterfullerene is lower than that of graphite.
(2) There are delocalized electrons in buckminsterfullerene.
(3) On complete combustion, buckminsterfullerene forms carbon dioxide and water.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(1) Molecules in buckminsterfullerene are bound by weak van der Waals' forces while graphite has a giant covalent structure. Thus, the melting point of buckminsterfullerene is lower than that of graphite.
(2) Each carbon atom has four outermost shell electrons. In a molecule of buckminsterfullerene, each carbon atom forms three single covalent bonds with other carbon atoms. The remaining electron is delocalized.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

17 Consider the following four fluorides:

CF_4, NF_3, PF_5 and SF_2

For each fluoride molecule,

Refer to page T15.

a) draw an electron diagram;

b) draw a 3-D structure, showing all electron pairs in the outermost shell of the central atom;

c) state the shape of the molecule.

18 'Nitrous oxide' gas, N_2O, is formed in the soil by denitrifying bacteria.

One model of the bonding in nitrous oxide includes a dative covalent bond between the oxygen atom and the central nitrogen atom. Complete the electron diagram for a molecule of nitrous oxide based on this model.

N N O

Suggest and explain a shape for the molecule.

Refer to page T15.

(OCR Advanced GCE, Chem. B (Salters), F335, Jun. 2010, 1(a)(ii))

19 Fluorine reacts with boron, B, to form the fluoride BF_3.

a) Suggest an equation for this reaction.
$2B(s) + 3F_2(g) \longrightarrow 2BF_3(g)$

b) Name the shape of a BF_3 molecule.
Shape: trigonal planar

Explain why BF_3 has this shape.
The three bonded pairs of electrons repel one another equally.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2010, 5(d))

20 Consider a dichlorine monoxide molecule (Cl_2O).
Refer to page T15.

a) Draw an electron diagram of the molecule, showing electrons in the *outermost shells* only.

b) Explain why the molecule does NOT have a linear shape.

21 Phosphorus pentachloride (PCl_5) is a chloride of phosphorus.
Refer to page T15.

a) Draw an electron diagram of a phosphorus pentachloride molecule. You should include only *outermost shell* electrons.

b) Draw and state the shape you would expect for the phosphorus pentachloride molecule.

c) Explain the shape of the phosphorus pentachloride molecule you have given in (b).

22 Both nitrogen and phosphorus are Group V elements. Phosphorus forms two chlorides, PCl_3 and PCl_5; but nitrogen forms only one chloride, NCl_3.

a) Suggest why NCl_5 does NOT exist.

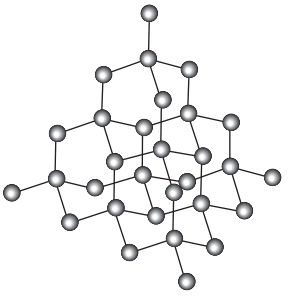
b) Draw the 3-D structures of PCl_3 and PCl_5.

(HKASLE, Paper 1, 2010, 4(a)–(b))

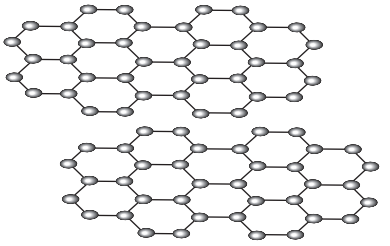
23 a) Diamond has a giant covalent structure. The carbon atoms are held together by strong covalent bonds.
A lot of heat is required to break the strong covalent bonds during melting.
b) Graphite has a layered structure. Weak van der Waals' forces exist between the layers.
The layers can easily slide over each other.
Hence graphite has a slippery feel.
c) Graphite has a layered structure. Within each layer, each carbon atom uses three outermost shell electrons in forming covalent bonds with three other atoms.
The remaining electron is delocalized between the layers of carbon atoms.
Graphite can conduct electricity due to the presence of delocalized electrons.
d) Only weak van der Waals' forces exist between the buckminsterfullerene molecules.
There is no need to break strong covalent bonds during melting.

23 Diamond and graphite are two naturally-occurring forms of carbon.

The diagrams below show the arrangement of the carbon atoms in diamond and in graphite. The black dots (●) represent carbon atoms.



diamond



graphite

a) Name the type of structure in diamond and explain, in terms of its bonding, why diamond has a high melting point.

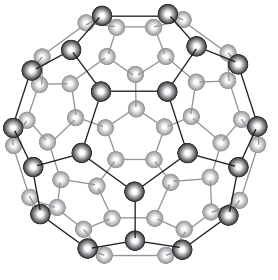
b) Explain, in terms of its structure, why graphite can act as a lubricant.

c) The structure of graphite has one feature in common with that of metals. This feature allows graphite to conduct electricity.

Suggest what this feature is and why it allows graphite to conduct electricity.

d) In 1985, a new form of carbon was discovered. It was called buckminsterfullerene after the architect Buckminster Fuller, who designed buildings with complex geometric shapes.

Buckminsterfullerene (C_{60}) has a simple molecular structure containing 60 carbon atoms per molecule. It looks a little bit like a football.



Suggest why buckminsterfullerene has a much lower melting point than diamond.

(Edexcel IGCSE, Chemistry, Paper 2C, Jun. 2011, 6)